

LAST UPDATED August 19, 2025

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	Honolulu, HI 96822	thomsonlab.org

PROFESSIONAL	Editor in Chief Elect	January 2025—Present
APPOINTMENTS	<i>Systematic Biology</i>	
	Journal of the Society of Systematic Biologists	
	Oxford University Press	

Professor	August 2022—Present
School of Life Sciences, College of Natural Sciences	
University of Hawai'i at Mānoa	

Graduate Program Chair	September 2020—Present
Ecology, Evolution, and Conservation Biology	
University of Hawai'i at Mānoa	

Research Associate	December 2022—Present
Department of Herpetology	
Natural History Museum of Los Angeles County	

Associate Director for Research	January 2021—July 2023
School of Life Sciences, College of Natural Sciences	
University of Hawai'i at Mānoa	

Associate Chair for Curriculum (interim)	August 2019—December 2019
Department of Biology, College of Natural Sciences	
University of Hawai'i at Mānoa	

Associate Professor	August 2017—July 2022
Department of Biology, College of Natural Sciences	
University of Hawai'i at Mānoa	

Graduate Faculty	May 2012—Present
Ecology, Evolution and Conservation Biology Graduate Program	
University of Hawai'i at Mānoa	

Graduate Faculty	January 2012—Present
Zoology Graduate Program	
University of Hawai'i at Mānoa	

Assistant Professor January 2012—July 2017
Department of Biology, College of Natural Sciences
University of Hawai'i at Mānoa

Postdoctoral Researcher October 2010—December 2011
Department of Evolution and Ecology
University of California, Davis

EDUCATION **Ph.D., Population Biology** September 2010
University of California, Davis

B.S. with High Honors, Biology May 2004
Rochester Institute of Technology

PUBLICATIONS
SUBMITTED OR
IN PRESS

69. Quijada, A, AJ Barley, P Rosen, IE Parra-Salazar, A Nieto-Montes de Oca, DH Leavitt, **RC Thomson**, T Reeder. *in review*. A new patternless morph population adds to the taxonomic intricacies within the *Aspidoscelis burti* complex. *Herpetological Review*

68. Barley, AJ, DV Ho, P Baumann, IJ Wang, HB Shaffer, RN Fisher, LN Gray, TJ Krabbenhoft, M Escalona, E Toffelmier, R Sahasrabudhe, O Nguyen, CW Fairbairn, E Beraut, **RC Thomson**. *in revision*. Speciation genomics in the tiger whiptail lizards (*Aspidoscelis tigris* complex). *Genome Biology and Evolution*.

67. Alvarez V, SR Fisher, GB Pauly, AJ Barley, AN Wright, RN Fisher, **RC Thomson**. *in revision*. Disentangling complex biological introduction histories: the ongoing invasion of the brown anole (*Anolis sagrei*).

66. Halpin-McCormick A, **RC Thomson**, RC Clarke, J Neyhart, MB Kantar. *in press*. Dissecting Genotype-Environment interactions with functional implications for parental selection in Cannabis Breeding. *Journal of Heredity*.

PUBLICATIONS -
GOOGLE
SCHOLAR

65. Toups BS, **RC Thomson**, JM Brown. 2025. Complex Models of Sequence Evolution Improve Fit, but not Gene Tree Discordance, for Tetrapod Mitogenomes. *Systematic Biology*. 74: 86-100

64. Alvarez V, RN Fisher, SA Hathaway, **RC Thomson**. *in review*. Widespread dispersal of a human commensal across the Pacific: Reconstructing the human-mediated invasion history of the house gecko (*Hemidactylus frenatus*). *Biological Invasions*. 27: 167

63. Barley AJ, A Nieto-Montes de Oca, NL Manriquez-Moran, **RC Thomson**. *in press*. Understanding Species Boundaries that Arise from Complex Histories: Gene Flow Across the Speciation Continuum in the Spotted Whiptail Lizards. *Systematic Biology*. 73: 901-919

62. Kusaka, CM, KL Utsumi, JE Cordes, AJ Barley, **RC Thomson**, JM Walker. 2024. *Aspidoscelis velox* (Plateau Striped Whiptail) in Jefferson County, Oregon, USA: Persistence of an introduced triploid parthenogenetic whiptail lizard. *Western Wildlife*. 11: 42-49
61. Wright, AN, AJ Barley, GB Pauly, LN Gray, **RC Thomson**. 2024. A historical resurvey of flatland lizards in the Mojave Desert. *Journal of Herpetology*. 58: 188-197
60. Fabreti LG, LM Coghill, **RC Thomson**, S Höhna, JM Brown. 2024. The Expected Behavior of Posterior Predictive Tests and its Unexpected Interpretation. *Molecular Biology and Evolution*. 41: msae051
59. Wishingrad V, **RC Thomson**. 2024. Testing concordance and conflict in spatial replication of landscape genetics inferences. *Molecular Ecology*. 33: e17104
58. Alvarez, V, S Fisher, AJ Barley, K Donmoyer, M Blom, **RC Thomson**, R Fisher. 2023. On the origin and current distribution of the Oceania snake-eyed skink (*Cryptoblepharus poecilopleurus*) in the Hawaiian Archipelago. *Pacific Science*. 77: 87-101.
57. Wishingrad, V, **RC Thomson**. 2023. Biogeographic inferences across spatial and evolutionary scales. *Molecular Ecology*. 32: 2055-2070.
56. Wishingrad V, **RC Thomson**. 2023. Temperate zone isolation by climate: an extension of Janzen’s 1967 hypothesis. *The American Naturalist*. 201: 302-314.
55. Cordes, JE, WM Walker, MA Paulissen, RC Thomson, AJ Barley. 2022. Longevity and Ontogeny of Color Pattern in Diploid Parthenogenetic *Aspidoscelis laredoensis* Clonal Complex A in Texas. *Sonoran Herpetologist*. 35:152-153
54. Barley AJ, A Nieto-Montes de Oca, NL Manriquez-Moran, **RC Thomson**. 2022. The evolutionary network of whiptail lizards reveals predictable outcomes of hybridization. *Science*. 377: 773-777
53. Parks R, SM Harrington, **RC Thomson**. 2022. Divergence dating and biogeography of Xenosauridae including fossils as terminal taxa. *Journal of Herpetology*. 56:349-354
52. **Thomson RC**, JM Brown. 2022. On the Need for New Measures of Phylogenomic Support. *Systematic Biology*. 71:917-920
51. Barley AJ, JE Cordes, JM Walker, **RC Thomson**. 2022. Genetic diversity and the origins of parthenogenesis in the teiid lizard *Aspidoscelis laredoensis*. *Molecular Ecology*. 31: 266-278
50. Barley A, TW Reeder, A Nieto-Montes de Oca, CJ Cole, **RC Thomson**. 2021. A new diploid parthenogenetic whiptail lizard from Sonora, Mexico is the ‘missing link’ in the evolutionary transition to polyploidy. *The American Naturalist*. 198:295-309

49. Ennen, J, M Agha, S Hazzard, W Matamoros, J Lovich, A Rhodin, J Iverson, **RC Thomson**, B Shaffer, C Hoagstrom. 2021. A watershed moment: focusing in on sub-basins to focus the geography of turtle conservation across the globe. *Biological Conservation*. 253: 108925
48. **RC Thomson**, PQ Spinks, HB Shaffer. 2021. A global phylogeny of turtles reveals a burst of climate-associated diversification on continental margins. *Proceedings of the National Academy of Sciences, USA*. 118(7): e2012215118
47. Harrington SM, V Wishingrad, **RC Thomson**. 2021. Properties of Markov Chain Monte Carlo performance across many empirical alignments. *Molecular Biology and Evolution*. 38: 1627-1640
46. Wishingrad V, **RC Thomson**. 2020. Ecological variability is associated with functional trait diversity in the western fence lizard (*Sceloporus occidentalis*). *Biological Journal of the Linnean Society*. 129: 414-424
45. Hime PM, AR Lemmon, EC Moriarty Lemmon, E Prendini, JM Brown, **RC Thomson**, JD Kratovil, BP Noonan, RA Pyron, PLV Peloso, ML Kortyna, JS Keogh, SC Donnellan, RL Mueller, CJ Raxworthy, K Kunte, SR Ron, S Das, N Gaitonde, DM Green, J Labisko, J Che, and DW Weisrock. 2021. Phylogenomics Reveals Ancient Gene Tree Discordance in the Amphibian Tree of Life. *Systematic Biology*. 70: 49-66
44. Campillo LC, Barley AJ, **Thomson RC**. 2019. Model-based species delimitation: are coalescent species reproductively isolated? *Systematic Biology*. 69: 708-721.
43. Pauly GB, Shaulsky MC, Barley AJ, Kennedy-Gold S, Stewart S, Keeney S, **Thomson RC**. 2020. Morphological Change during Rapid Population Expansion Confounds Leopard Frog Identifications in the Southwestern United States. *Copeia*. 108: 299-308.
42. Campillo LC, JD Manthey, **RC Thomson**, PA Hosner, RG Moyle. 2020. Genomic differentiation in an endemic Philippine genus (Aves: *Sarcophanops*) due to geographic isolation on recently disassociated islands. *Biological Journal of the Linnean Society*. 131: 814-821.
41. Barley AJ, A Nieto-Montes de Oca, TW Reeder, NW Manríquez-Morán, JC Arenas Monroy, O Gallegos Hernández, **RC Thomson**. 2019. Complex patterns of hybridization and introgression across evolutionary timescales in Mexican whiptail lizards. *Molecular Phylogenetics and Evolution*. 132: 284-295.
40. Gray, LN, AJ Barley, S Poe, **RC Thomson**, A Nieto-Montes de Oca, IJ Wang. 2019. Phylogeography of a widespread lizard complex reflects patterns of both geographic and ecological isolation. *Molecular Ecology*. 28: 644-657.

39. Riggs RA, AD Taylor, **RC Thomson**, RH Cowie. 2019. Sexual Dimorphism and Seasonal Variability of Shield Size in the Endangered Hawaiian Coot, *Fulica alai*. *Waterbirds*. 42: 70-77.
38. Brown JM, **RC Thomson**. 2018. Evaluating Model Performance in Evolutionary Biology. *Annual Review of Ecology, Evolution, and Systematics*. 49: 95-114
37. Vinnikov KA, **RC Thomson**, TA Munroe. 2018. Revised classification of the Righteye Flounders (Teleostei: Pleuronectidae) based on multilocus phylogeny with complete taxon sampling. *Molecular Phylogenetics and Evolution*. 125: 147-162
36. Richards, EM, JM Brown, AJ Barley, RA Chong, **RC Thomson**. 2018. Unexpected variation across mitochondrial gene trees and evidence for systematic error: How much gene tree variation is biological? *Systematic Biology*. 67: 847-860
35. Höhna S, LM Coghill, GG Mount, **RC Thomson**, JM Brown. 2018. P³: Phylogenetic posterior prediction in RevBayes. *Molecular Biology and Evolution*. 35: 1028-1034
34. Brown JM, **RC Thomson**. 2018. The behavior of Metropolis-coupled Markov chains when sampling rugged (phylogenetic) distributions. *Systematic Biology*. 67: 729-734
33. Barley AJ, JM Brown, **RC Thomson**. 2018. Impact of model violations on the inference of species boundaries under the multispecies coalescent. *Systematic Biology*. 67: 269-284
32. **RC Thomson**, PQ Spinks, HB Shaffer. 2018. Molecular phylogeny and divergence of the Map Turtles (Emydidae: *Graptemys*). *Molecular Phylogenetics and Evolution*. 121: 61-70.
31. Brown JM, **RC Thomson**. 2017. Bayes factors unmask highly variable information content, bias, and extreme influence in phylogenomic analyses. *Systematic Biology*. 66: 517-530.
30. Nieto-Montes de Oca A, AJ Barley, RN Meza-Lázaro, UO García-Vázquez, JG Zamora-Abrego, **RC Thomson**, AD Leaché. 2017. Phylogenomics and species delimitation in the knob-scaled lizards of the genus *Xenosaurus* (Squamata: Xenosauridae) using ddRADseq data reveal a substantial under-estimation of diversity. *Molecular Phylogenetics and Evolution*. 106: 241-253
29. Barley AJ, **RC Thomson**. 2016. Assessing the performance of DNA barcoding using posterior predictive simulations. *Molecular Ecology*. 25:1930-1943
28. Spinks PQ, **RC Thomson**, E McCartney-Melstad, HB Shaffer. 2016. Phylogeny and temporal diversification of the New World pond turtles (Emydidae). *Molecular Phylogenetics and Evolution*. 103:85-97

27. Dong CM, TN Engstrom, **RC Thomson**. 2016. Origins of softshell turtles in Hawaii with considerations for conservation. *Conservation Genetics*. 17:207-220
26. Barley AJ, PJ Monnahan, **RC Thomson**, LL Grismer, RM Brown. 2015. Sun skink landscape genomics: assessing how microevolutionary processes shape genetic and morphological diversity across a heterogeneous and fragmented landscape. *Molecular Ecology*. 24:1696-1712
25. **Thomson RC**, DC Plachetzki, DL Mahler, BR Moore. 2014. A critical appraisal of the use of microRNA data in phylogenetics. *Proceedings of the National Academy of Sciences, USA*. 111:E3659-E3668
24. Long EC, **RC Thomson**, AM Shapiro. 2014. A time-calibrated phylogeny of the butterfly tribe Melitaeini. *Molecular Phylogenetics and Evolution*. 79:69-81
23. Spinks PQ, **RC Thomson**, M Gidis, HB Shaffer. 2014. Multilocus phylogeny of the new-world mud turtles (Kinosternidae) supports the traditional classification of the group. *Molecular Phylogenetics and Evolution*. 76:254-260
22. Spinks PQ, **RC Thomson**, HB Shaffer. 2014. The advantages of going large: genome-wide SNPs clarify the complex population history and systematics of the threatened western pond turtle. *Molecular Ecology*. 23:2228-2241
21. Shaffer HB, P Minx, DE Warren, AM Shedlock, **RC Thomson**, N Valenzuela, J Abramyan, D Badenhorst, KK Biggar, GM Borchert, CW Botka, RM Bowden, EL Braun, AM Bronikowski, BG Bruneau, LT Buck, B Capel, TA Castoe, M Czerwinski, KD Delehaunty, SV Edwards, CC Fronick, MK Fujita, L Fulton, TA Graves, RE Green, W Haerty, R Hariharan, LW Hillier, AK Holloway, D Janes, FJ Janzen, C Kandath, L Kong, APJ de Koning, Y Li, R Literman, SE McGaugh, L Mork, M O’Laughlin, RT Paitz, DD Pollock, CP Ponting, S Radhakrishnan, BJ Raney, JM Richman, J St. John, T Schwartz, A Sethuraman, PQ Spinks, KB Storey, N Thane, T Vinar, LM Zimmerman, WC Warren, ER Mardis, and RK Wilson. 2013. The western painted turtle genome, a model for the evolution of extreme physiological adaptations in a slowly evolving lineage. *Genome Biology*. 14: R28
20. Spinks PQ, **RC Thomson**, GB Pauly, CE Newman, G. Mount, HB Shaffer. 2013. Misleading phylogenetic inferences based on single-exemplar sampling in the turtle genus *Pseudemys*. *Molecular Phylogenetics and Evolution*. 68:269-281
19. Lambert MR, SN Nielsen, AN Wright, **RC Thomson**, HB Shaffer. 2013. Microhabitat characteristics favor introduced red-eared sliders over threatened western pond turtles in a human-dominated landscape. *Chelonian Conservation and Biology*. 12: 192-199

18. Spinks PQ, **RC Thomson**, Y Zhang, J Che, Y Wu, HB Shaffer. 2012. Species boundaries and phylogenetic relationships in the critically endangered Asian box turtle genus *Cuora*. *Molecular Phylogenetics and Evolution*. 63: 656-667
17. Spinks PQ, **RC Thomson**, B Hughes, B Moxley, RM Brown, A Deismos, HB Shaffer. 2012. Cryptic variation and the tragedy of unrecognized taxa: the case of international trade in the spiny turtle *Heosemys spinosa* (Testudines: Geoemydidae). *Zoological journal of the Linnean Society*. 164: 811-824
16. Holzman R, DC Collar, SA Price, CD Hulseay, **RC Thomson**, PC Wainwright. 2012. Biomechanical trade-offs bias rates of evolution in the feeding apparatus of fishes. *Proceedings of the Royal Society – Series B*. 279: 1287-1292
15. Johnson JR, **RC Thomson**, SJ Micheletti, HB Shaffer. 2011. The origin of tiger salamander populations in California: Introductions or relicts? *Conservation Genetics*. 12: 355-370
14. **Thomson RC**, PQ Spinks HB Shaffer. 2010. Distribution and abundance of exotic red-eared sliders (*Trachemys scripta elegans*) in California's Sacramento River basin and possible impacts on native western pond turtles (*Emys marmorata*). *Chelonian Conservation and Biology*. 9: 297-302
13. **Thomson RC**, HB Shaffer. 2010. Rapid progress on the vertebrate tree of life. *BMC Biology*. 8: 19
12. **Thomson RC**, IJ Wang, JR Johnson. 2010. *Invited Review*: Genome-enabled development of molecular markers for ecology, evolution, and conservation. *Molecular Ecology*. 19: 2184-2195
11. Spinks PQ, **RC Thomson**, AJ Barley, HB Shaffer. 2010. Testing avian, squamate, and mammalian nuclear markers for cross amplification in turtles. *Conservation Genetics Resources*. 2:127–129
10. Barley AJ, PQ Spinks, **RC Thomson**, HB Shaffer. 2010. Fourteen nuclear genes provide phylogenetic resolution for difficult nodes in the turtle tree of life. *Molecular Phylogenetics and Evolution*. 55: 1189-1194
9. **Thomson RC**, HB Shaffer. 2010. Sparse Supermatrices for Phylogenetic Inference: Taxonomy, Alignment, Rogue Taxa, and the Phylogeny of Living Turtles. *Systematic Biology*. 59: 42-58
8. Spinks PQ, **RC Thomson**, HB Shaffer. 2010. Nuclear gene phylogeography reveals the historical legacy of an ancient inland sea on lineages of the western pond turtle, *Emys marmorata* in California. *Molecular Ecology*. 19: 542-556
7. **Thomson RC**. 2009. PhyLIS: a simple Gnu/Linux distribution for phylogenetics and phyloinformatics. *Evolutionary Bioinformatics*. 5: 91-95

6. Spinks PQ, **RC Thomson**, GA Lovely and HB Shaffer. 2009. Assessing what is needed to resolve a molecular phylogeny: simulations and empirical data from emydid turtles. *BMC Evolutionary Biology*. 9: 56
5. Spinks PQ, **RC Thomson**, HB Shaffer. 2009. A reassessment of *Cuora cyclornata* Blanck, McCord and Le, 2006 (Testudines, Geoemydidae) and a plea for taxonomic stability. *Zootaxa*. 2018: 58-68
4. **Thomson RC**, AM Shedlock, SV Edwards, HB Shaffer. 2008. Developing markers for multilocus phylogenetics in non-model organisms: a test case with turtles. *Molecular Phylogenetics and Evolution*. 49: 514-525
3. Iverson JB, RM Brown, TS Akre, TJ Near, M Le, **RC Thomson**, DE Starkey. 2007. In search of the tree of life for turtles. in Defining Turtle Diversity: Proceedings of a workshop on genetics, ethics, and taxonomy for tortoises and freshwater turtles. HB Shaffer, NN Fitzsimmons, A Georges, AGJ Rhodin , Eds. *Chelonian Research Monographs* 4: 85-106
2. Shaffer HB and **RC Thomson**. 2007. Delimiting species in recent radiations. *Systematic Biology* 56: 896-906

BOOKS

1. **Thomson RC**, AN Wright, HB Shaffer. 2016. California amphibian and reptile species of special concern. *University of California Press*. Oakland, CA. 390 p.

JOURNAL SPECIAL ISSUES

- Thomson RC** and JM Brown (eds.). 2022. On the need for new measures of phylogenomic support. *Systematic Biology*. 71(4)

OTHER PUBLICATIONS

- Thomson RC**. Here be (Gila) monsters. *Desert Oracle*. Spring 2016: 17-19

GRANTS

- 2022 University of Hawaii Undergraduate Research Opportunities Program: Faculty Mentoring Grant: Determining the provenance and conservation needs of extralimital six-lined racerunner lizards using genetics. \$5,000.
- 2021-2023 Ocean Park Conservation Foundation, Hong Kong: Assessing the genetic diversity and ecological role of the endangered wattle-necked softshell turtle (*Palea steindachneri*) through the study of remnant populations. (PI: Jonathan Fong, Co-PIs: Yik-Hei Sung, Duc Minh Le, Robert Thomson). HK\$297,700 (approx. \$38,234 US)
- 2020-2024 NSF OPUS: CRS: Collaborative Research: A Synthetic View of Evolutionary Heterogeneity and the Tree of Life (PI: Robert Thomson, PI: Jeremy Brown). \$215,223
- 2020-2021 Extreme Science and Engineering Discovery Environment (XSEDE) DEB190017: Computational tools for understanding systematics, hybridization, incomplete coalescence, and sex system reversal in whiptail lizards. (PI: Robert Thomson, Co-PI: Anthony Barley). 7,639,010 Service Units, 1.12 TB storage (XSEDE-calculated value \$398,910)

- 2018-2023 NSF DEB Collaborative Research: Species delimitation, hybridization and the origin of parthenogenesis in Whiptail lizards (*Aspidoscelis*) (PI: Robert Thomson, Co-PI: Tod Reeder). \$752,179
- 2015-2018 Arnold and Mabel Beckman Foundation Postdoctoral Fellowship (Postdoctoral Fellow: Anthony Barley, Mentor: Robert Thomson) \$184,172
- 2014-2017 NSF DBI Collaborative Research: ABI Innovation: A Bayesian Evolutionary Tree Analysis Database (PI: Brian Moore, PI: Robert Thomson). \$549,728
- 2014-2017 NSF DEB Collaborative Research: Bayesian model checking for phylogenetics in the post-genomic era (PI: Jeremy Brown, Co-PI: Robert Thomson). \$600,208
- 2008 California Department of Fish and Game “Determining Management Units for the Western Pond Turtle (*Emys marmorata*)” \$118,575
(Senior Investigator with HB Shaffer and PQ Spinks)
- 2008 California Department of Fish and Game “Amphibian and Reptile Species of Special Concern in California” \$96,297
(Senior Investigator with HB Shaffer)
- 2008 UC Davis graduate student award in engineering and computer science \$8,106
- 2008 UC Davis Biological invasions NSF IGERT research award “The impact of exotic turtles on an endangered native: Melding research and education” \$500
- 2007 Society of Systematic Biologists Graduate Student Award “Genomic perspective on systematics of the map turtles (*Graptemys*)” \$1,650
- 2007 NSF DEB Doctoral Dissertation Improvement Grant: Systematics of rapid radiations: New approaches in the post-genomic age. \$12,000
- 2005-2007 UC Davis competitive grants in support of research and travel, 6 awards \$5,025 total
- 2003 Rochester Academy of Science undergraduate research grant \$500

AWARDS & HONORS

- 2015 University of Hawaii Regent’s Medal For Excellence in Research
- 2010 UC Davis Merton Love Award for Outstanding Dissertation in Evolution and Ecology
- 2004 Rochester Institute of Technology Baldwin Award for Outstanding Undergraduate in the Life Sciences

INVITED
SEMINARS &
SYMPOSIA

Times Higher Education Teaching Excellence Summit	November 2021
Panel Discussion “Innovation in STEM Education: Lessons from the past year”	
Association of Pacific Rim Universities	October 2020
Panel Discussion “STEM Education across the APRU Network: Shaping Learning Experiences for Students”	
Evolution Conference	June 2019
Society of Systematic Biology Spotlight “The Bright Side of Phylogenetics”	
Evolution Conference	June 2017
Society of Systematic Biology Spotlight “Uncertainty in the face of strong support: new approaches and examples in the identification of signal, noise, and conflict in phylogenetic datasets”	
Hawaii Institute of Marine Biology	February 2017
Symposium on Conservation Genetics	
Louisiana State University Museum of Natural Science	September 2016
Society for Study of Amphibians and Reptiles	July 2015
Symposium “Frontiers in Integrative Organismal Biology: Herpetological Horizons”	
Hawaii Institute of Marine Biology	October 2012
University of Hawaii. Department of Biology	March 2011
University of New Mexico. Department of Biology	October 2010
Merton Love Award Seminar. UC Davis	June 2010
Evolution Conference	June 2009
Symposium “Ernst Mayr Student Research Symposium”	
CSU, Chico. Department of Biological Sciences	October 2008
Society for Integrative and Comparative Biology	January 2008
Symposium: “Reptile genomics and evolutionary genetics”	
Powdermill freshwater turtle conference	August 2006
Evolution Conference	June 2006
Symposium: “Species delimitation: new approaches for discovering diversity”	
Museum of Comparative Zoology, Harvard University	August 2005
Symposium: “Defining Turtle Diversity”	

CONTRIBUTED
PRESENTATIONS
& POSTERS
(LAST 10 YEARS
ONLY)

Barley, AJ, A Nieto-Montes de Oca, NL Manriquez-Moran, **RC Thomson**. Speciation, hybridization, and reproductive mode evolution in whiptail lizards (*Aspidoscelis*). 2023 Society of Systematic Biologists Standalone Conference. Mexico City, Mexico.

Alvarez V, **RC Thomson**. Lizards in paradise: How the brown anole made its way to Hawai'i and Southern California. 2022 Joint Meeting of Ichthyologists and Herpetologists.

Barley AJ, A Nieto-Montes de Oca, NL Manriquez-Moran, **RC Thomson**. Species boundaries and pervasive gene flow in the spotted whiptail lizards (*Aspidoscelis gularis* complex). 2022 Joint Meeting of Ichthyologists and Herpetologists - Biology of Whiptails Symposium.

Barley AJ, A Nieto-Montes de Oca, NL Manriquez-Moran, **RC Thomson**. The evolutionary network of whiptail lizards reveals predictable outcomes of hybridization. 2022 Joint Meeting of Ichthyologists and Herpetologists - Biology of Whiptails Symposium.

Alvarez V, S Fisher, Aj Barley, RN Fisher, **RC Thomson**. Hawaii's only native land reptile may be in need of conservation. 2022 Hawaii Conservation Conference.

Fabreti LG, LM Coghill, **RC Thomson**, S Höhna, JM Brown. Posterior predictive tests have surprising behavior, but reveal important idiosyncratic patterns of molecular evolution. 2022 Evolution.

RC Thomson. Reptiles and Amphibians of O'ahu. 2022 City Nature Challenge at the Bishop Museum.

RC Thomson. A Global Phylogeny of Turtles Reveals a Burst of Climate-associated Diversification on Continental Margins. 2020 Turtle Survival Alliance.

LC Campillo, E VanderWerf, **RC Thomson**. Population genomics of an endemic Hawaiian bird and the effect of habitat fragmentation on 'elepaio demography. Poster. 2019 Evolution.

V Wishingrad, **RC Thomson**. Spatial scale and gene flow in landscape genetics processes. 2019 Evolution.

RC Thomson, JM Brown, R Lanfear. How should we assess support in phylogenomic studies? Panel Discussion. 2019 Evolution.

S Harrington, **RC Thomson**. Patterns of phylogenetic MCMC convergence across empirical datasets. 2019 Evolution.

AJ Barley, **RC Thomson**, A Nieto Montes de Oca, N Manríquez Morán. Reconstructing patterns of hybridization and the origins of parthenogenesis in whiptail lizards (*Aspidoscelis*). 2019 Evolution.

JM Brown, **RC Thomson**, L Coghill, A Schoonmaker, L Rodriguez. The role of model testing and evaluation in phylogenomics. 2019 Evolution.

- S Harrington, R Parks, **RC Thomson**. Tip-dated phylogenetics and biogeography of *Xenosaurus*. 2019 JMIH.
- V Wishingrad, **RC Thomson**. *Sceloporus occidentalis* lizard landscape genetics in the Sierra Nevada mountain range. 2019 JMIH.
- RC Thomson**. Detecting causes of conflict in phylogenomic datasets. Phylogenetics and Math Symposium, University of Hawaii
- Shaffer HB, **RC Thomson**, J Vu. A global phylogeny of turtles and how it informs conservation planning. 2017 Evolution
- Campillo LC, AJ Barley, **RC Thomson**. Using a classic speciation data set to test coalescent methods for species delimitation. 2017 Evolution
- Coghill L, A Schoonmaker, L Rodriguez, **RC Thomson**, JM Brown. The effect of heterotachy on phylogenomics. 2017 Evolution
- Barley AJ, A Nieto-Montes de Oca, **RC Thomson**. Elucidating the evolutionary history of Mexican whiptail lizards. 2017 Evolution
- Kliman RM, LS Maroja, SA Price, JJ Smith, **RC Thomson**. EvoED Digital Library: SSE's education resource. Poster. 2016 Evolution
- Brown JM, **RC Thomson**. Climbing peaks and crossing valleys: Metropolis coupling and rugged phylogenetic distributions. 2016 Evolution
- Hime P, +21 co-authors. Phylogenomics of Amphibia and the Nature of Support and Signal in Big Data Sets. 2016 Evolution
- Coghill LM, Brown JM, **RC Thomson**. Big data and outlier loci: A cautionary tale with genome-scale phylogenetic data. 2016 Evolution
- Barley AJ, **RC Thomson**. Species delimitation's new conceptual roots. 2016 Evolution
- Richards EJ, AJ Barley, JM Brown, RA Chong, **RC Thomson**. An empirical assessment of model performance tests in phylogenetics. 2016 Evolution
- Thomson RC**, PQ Spinks, HB Shaffer, AJ Barley. Phylogeny of the Map Turtles (*Graptemys*). 2016 Joint Meeting of Ichthyologists and Herpetologists
- Wishingrad V, AJ Barley, **RC Thomson**, AN Wright. Evidence-based threat assessment: lessons from California herpetofauna. Poster. 2016 Joint Meeting of Ichthyologists and Herpetologists
- Nieto-Montes de Oca, A, AJ Barley, RN Meza-Lázaro, UO García-Vázquez, JG Zamora-Abrego, **RC Thomson**, AD Leaché. Phylogenomics of the knob-scaled lizards (*Xenosaurus*). 2016 Joint Meeting of Ichthyologists and Herpetologists
- AN Wright, GB Pauly, AJ Barley, **RC Thomson**. Historical Resurveys of Lizard Communities in the Mojave Desert. 2016 Joint Meeting of Ichthyologists and Herpetologists

Dong CM, T Engstrom, **RC Thomson**. Origins of softshell turtles in Hawaii with considerations for conservation. 2015 Society for Study of Amphibians and Reptiles

Brown JM, **RC Thomson**. Comparing the amount and quality of information from different phylogenomic sequencing strategies: a case study with amniotes. 2015 Evolution

Barley AJ, **RC Thomson**. Substitution models and genetic distances: posterior predictive assessments of model adequacy in DNA barcoding. 2015 Evolution

TEACHING

Instructor

ZOOL750 Fall 2019, Spring 2024
Models and Inference in Conservation

BIOL470 every fall, 2023—Present
Evolutionary Biology

ZOOL780 & ZOOL781 alternate years, Spring 2015—Present
Foundations of Evolution and Ecology I & II

BOT/ZOOL670 alternate years, Spring 2014—Present; Fall 2020
Scientific Teaching Tools for Active Learning

ZOOL690 Fall 2021
Conservation Biology

ZOOL480 every Spring, 2012—2019
Evolutionary Biology

ZOOL171 Fall 2018
Introductory Biology

ZOOL691B (seminar) Fall 2017, Fall 2018, Spring 2021, Spring 2022, Fall 2022
Phylogenetics and Speciation

ZOOL691B (seminar) Fall 2015
Bioinformatics Data Skills

ZOOL691B (seminar) Spring 2013
Statistical Phylogenetics: Problems, Prospects and Practice

Workshop Instruction

Workshop in Phylogenomics (instructor and organizer) 2020, 2024
Transmitting Science Course (Online)

Workshop in Phylogenomics (instructor and organizer)	2019
Transmitting Science Course in Valencia, Spain	
Workshop in Applied Phylogenetics	2007—2015, 2017, 2019
Held at Bodega Marine Lab, Bodega Bay, CA	
Phylogenomics workshop for Edwin W. Pauley Program	2013
Held at Hawaii Institute for Marine Biology, Kaneohe, HI	
Workshop in High Performance Computing for Phylogenetics	2010
Held at NIMBioS, Knoxville, TN	

SERVICE

Editing and Reviews

<i>Editor-in-Chief Elect</i> , Systematic Biology	2025
<i>Associate Editor</i> , Systematic Biology	2017—2024
<i>Editorial Board</i> , Horizons (UH undergrad research journal)	2020—Present
<i>Editorial Board</i> , Frontiers in Ecology and Evolution	2021—2022
<i>Editorial Board</i> , SSE EvoEd Digital Library	2016—2019

Reviewer for Journals

American Naturalist, Aquatic Invasions, Asian Herpetological Research, Biological Conservation, Biology Letters, BMC Bioinformatics, Briefings in Bioinformatics, Chelonian Conservation & Biology, Conservation Biology, Conservation Genetics, Conservation Genetics Resources, Diversity and Distributions, Ecology, Ecology and Evolution, Endangered Species Research, Frontiers in Ecology and Evolution, Genetics, Herpetologica, Herpetological Review, Journal of Evolutionary Biology, Journal of Herpetology, Journal of Molecular Evolution, Methods in Ecology and Evolution, Molecular Ecology, Molecular Ecology Resources, Molecular Phylogenetics & Evolution, Northwestern Naturalist, Organisms: Diversity & Evolution, Phyllomedusa, PLoS One, Proceedings of the National Academy of Sciences USA, Proceedings of the Royal Society B, Systematic Biology, Zoology

Reviewer for agencies, funders, and other organizations

Austrian Science Fund, California Department of Fish & Wildlife, Czech Science Foundation, Marsden Fund (Royal Society of New Zealand), Research Committee of Lingnan University (Hong Kong), Swiss National Science Foundation, US National Science Foundation, University of California Press, US Fish & Wildlife Service

Symposia and Event Organization

<i>Machine learning and new inference algorithms: expanding what is possible in evolutionary biology and phylogenetic analysis</i>	2025
Society of Systematic Biologists symposium at Evolution Conference in Athens, GA	

<i>The Biology of Whiptail Lizards (Aspidoscelis)</i>	2022
Joint Meeting of Ichthyologists and Herpetologists in Spokane, WA	
<i>City Nature Challenge at the Bishop Museum</i>	2022
Public presentations and Bioblitz, Honolulu, HI	
<i>New Measures of Phylogenetic Support for the Genomic Era</i>	2019
Society of Systematic Biologists symposium at Evolution Conference in Providence, RI	
<i>Phylogenetics and Math</i>	2019
Special Symposium ahead of American Mathematical Society regional meeting in Honolulu	
<i>Comparing phylogenetic trees: Why and How?</i>	2018
Symposium at Evolution Conference in Montpellier, France	
<i>Tester Symposium</i> Planning, UHM Biology	2013—2015

Service and Community

UHM College of Nat. Sci. Dean's Research Council	2024—present
HDOA Advisory Subcommittee on Land Vertebrates	2023—present
Society of Systematic Biologists Council (elected)	2023—2025
Organizer, City Nature Challenge Oahu	2020—2024
SSE GSAC Student-Faculty Mentoring Program	2023, 2024
UH SoLS Faculty Mentoring (two junior faculty mentees)	2020
Interview, The Guardian Science Weekly Podcast	2021
Expert Advice for news outlets and writers	2014, 2015, 2017, 2020, 2025
Editor, TreeThinkers blog	2013—2018
Scientist Participant, Honolulu Museum of Art School, ARTSCI	2016
Event Coordinator, California State Science Olympiad	2009
Web Admin, Bodega Bay Phylogenetics Wiki	2008—2013
Protocol Author, RIT Center for Bioscience Edu. and Tech.	2004
Team Leader, Great Lakes Basin Marsh Monitoring Program	2001—2004

Certifications and trainings

Wilderness First Responder	2023—2025
UH DEI Search Advocate Program	2021—2024
LGBTQ+ Safe Zone Program	2018
HHMI & NAS Summer Institute on Scientific Teaching	2013, 2016

PROFESSIONAL SOCIETIES & MEMBERSHIPS

Society of Systematic Biologists (SSB)
 Society for the Study of Evolution (SSE)
 Society for the Study of Amphibians and Reptiles (SSAR)
 IUCN Species Survival Commission—Tortoise and Freshwater Turtle Specialist Group (IUCN—TFTSG)
 National Institute on Scientific Teaching (NIST)
 American Association for the Advancement of Science (AAAS)

Postdoctoral

Dr. Anthony Barley (2014-2021), Dr. Sean Harrington (2017-2019)

Graduate Students

Valentina Alvarez (PhD 2024); Luke Campillo (PhD 2022); Caroline Dong (MS 2015); Karina Mareno (PhD current); Natalie Myers (MS 2023); Emilie Richards (MS 2016); Van Wishingrad (PhD 2021)

Graduate Student Committees

Kyhl Austin (PhD current); Jacob Buehler (MS 2016); Jared Bernard (MS current); Sean Canfield (PhD 2021); Richard Coleman (PhD 2019); Emily Conklin (PhD 2024); Stefan Cranston (MS current); Molly Hagemann (MS current); Mike Henley (PhD 2021); Derek Kraft (PhD 2021); Aki Laruson (PhD 2018); Claire Lewis (PhD 2022); Jessica Maxfield (PhD 2019); Ikenna Nometa (PhD 2025); Brad Reil (PhD current); Randi Riggs (MS 2016); Julio Rivera (PhD 2017); Michael San Jose (PhD 2018); Robyn Screen (PhD 2022); Helen Sung (PhD 2025); Kaho Tisthammer (PhD 2017); Kirill Vinnikov (PhD 2019); Ellen Waldrop (MS 2013); Christie Wilcox (PhD 2014); Melissa Van Kleek (PhD 2015)

Undergraduate Students

Hana Blatter (UH 2014); Justin Chan (UH current); Sam Fisher (UH 2015); Max Lambert (UCD 2010); Ashlee Lum (UH 2022); Ann Marsolais (UH 2015); Quinn Moon (UH 2023); Sharell Nielsen (UCD 2010); Evan Padro (UH 2015); Riley Parks (UH 2021); Melissa Seymour (UH 2018); Maya Shaulsky (UH 2018); Raon Traynor (UH current); Preston Varley (UH 2017); Dominique Wenzler-Steves (UH 2022); Ian Wright (UCD 2007)